Table 1: Key ratios. The bound requires $\delta \geq C\,\rho \ln(1/\rho)$ for a constant $C > 0$: sparser codes (smaller ρ) and higher observation dimension (larger δ) make recovery easier.

Symbol	Definition	Interpretation
ρ	$\frac{k}{d_h}$	**Sparsity.** Fraction of non-zeros in $\mathbf{h}$.
δ	$\frac{d_y}{d_h}$	**Undersampling.** Ratio of observation dimension d_y to code dimension d_h. $\delta < 1$ is the overcomplete regime.

per-sample to fully amortised inference, plus a linear-probe baseline. All are evaluated on both in-distribution (ID) and out-of-distribution (OOD) test sets, where the OOD split withholds specific combinations of active latents during training (Figure 2).

We consider latent variables $\mathbf{z} \in [0,1]^{d_z}$ with at most k non-zero entries, each sampled uniformly on $[0,1]$ when active, observed through a linear mixing $\mathbf{y} = \mathbf{A}\mathbf{z}$, where $\mathbf{A} \in \mathbb{R}^{d_y \times d_z}$ with $d_y < d_z$. Details in Section A. Two ratios govern recovery difficulty (Table 1): sparsity $\rho = k/d_z$ and undersampling $\delta = d_y/d_h$. We evaluate these using the metrics below. The target $t = \mathbf{1}\{z_1 > 0.5\}$ is predicted from inferred codes $\mathbf{h}$.

Metrics. We report three quantities on both ID and OOD data (details in Section D.1). The *mean correlation coefficient* (*MCC*) (Hyvarinen and Morioka, 2016) evaluates identifiability s.t. MCC = 1 for a representation identified up to permutation and rescaling. *Accuracy* trains a logistic probe on the inferred codes $\mathbf{h}$ to predict the binary label t, applying the *same* supervised classifier to every method's codes, isolating the effect of the representation. *AUC* is computed per-feature without a trained classifier: for each code dimension, we use the raw activation as a score and compute ROC AUC; the single best feature on ID data is selected and its AUC is reported on both splits. AUC tests whether the label is isolated in an individual feature—a stronger condition than accuracy, which can exploit combinations of features. In the main text we report MCC (identifiability) and Accuracy (for fair downstream comparison between an unsupervised and a supervied learning method). AUC and additional metrics are in the appendix.

Methods. We compare methods spanning four families: *sparse coding* (per-sample ℓ_1 inference with oracle or learned dictionaries), *SAEs*, *frozen decoder* (per-sample FISTA on a frozen SAE-learned dictionary), and *refined hybrids* (FISTA warm-started from SAE codes). The last two families disentangle dictionary quality from inference: frozen-decoder methods replace only the encoder while reusing the SAE's dictionary, whereas refined methods additionally test whether the SAE's output provides a useful initialisation. A linear-probe baseline (or, a supervised skyline) operates directly on $\mathbf{y}$. Architectural and optimisation details are in C.1. All SAEs (ReLU (Cunningham et al., 2023), JumpReLU (Rajamanoharan et al., 2024), Top-K (Gao et al., 2024), MP (Costa et al., 2025)) share the same decoder dimension and are trained with identical optimiser settings, differing only in the activation function governing the encoder's sparsity mechanism.

4.1 THE AMORTISATION GAP PERSISTS ACROSS UNDERSAMPLING RATIOS

Compressed sensing theory predicts a phase transition in sparse recovery: once the number of observations d_y exceeds $O(k \ln(d_z/k))$, per-sample ℓ_1 methods recover the latent code exactly. We sweep the undersampling ratio $\delta = d_y/d_h$ across a grid of latent dimensions $d_z \in \{50, 100, 200\}$ and sparsity levels $k \in \{3, 5, 10\}$, and report ID MCC (Figure 5). Phase transition with other metrics are reported in Section D. If SAEs solved the same sparse inference problem, they would exhibit the same transition and reach the same asymptotic performance. A persistent gap between the two would reveal the cost of amortising inference into a fixed encoder.

Per-sample methods exhibit the predicted phase transition. FISTA (oracle) transitions sharply to near-perfect MCC once δ passes a critical threshold, and the transition sharpens with d_z, matching the theoretical prediction. The empirical transition point is consistent with the constant $C \approx 2$ used in the compressed-sensing bound $\delta \geq C\rho \ln(1/\rho)$, validating the bound's practical relevance. DL-FISTA follows the same curve, shifted right by the cost of learning the dictionary. SAE variants also improve with δ—they are not insensitive to the undersampling ratio—but plateau at 0.2–0.5 MCC, well below the near-perfect recovery that per-sample methods achieve in the same regime. Crucially, the gap does not close at high δ, implying that even when the problem is well within the regime where compressed sensing guarantees exact recovery, the amortised encoder remains the bottleneck.

Takeaway. Both SAEs and sparse coding benefit from less aggressive undersampling (higher δ), but SAEs saturate far below per-sample methods.

4.2 SCALING UP LATENT DIMENSION DOES NOT CLOSE THE COMPOSITIONAL GAP

All unsupervised methods face a more challenging problem as d_z grows, as the dictionary has more columns and the space of sparse patterns expands combinatorially. We hypothesise that per-sample methods should degrade more gracefully if the bottleneck is dictionary learning, whereas SAEs should degrade faster if the bottleneck is amortised inference.

We sweep $d_z \in \{50, 100, 500, 1\text{K}, 5\text{K}, 10\text{K}\}$ with k and δ held fixed and report ID MCC (Figure 6) and OOD accuracy (Figure 7). As d_z grows, dictionary learning becomes harder

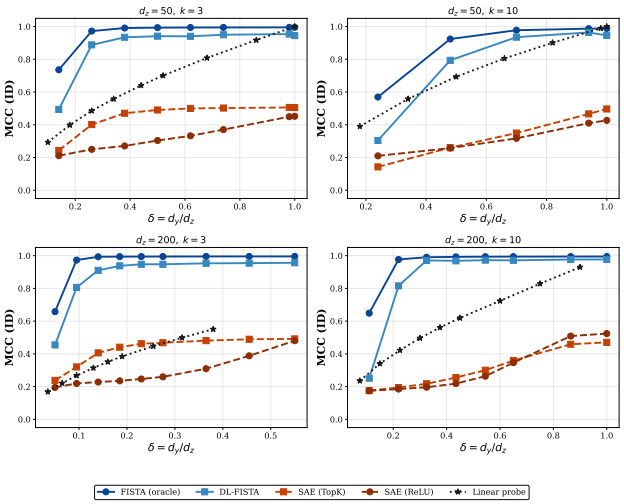

Figure 5: **The amortisation gap persists across under-sampling ratios.** Per-sample methods (FISTA) exhibit a sharp phase transition to near-perfect MCC once δ exceeds the compressed-sensing threshold; SAEs plateau at 0.2–0.5 MCC regardless of δ. Each panel shows a different (d_z, k) combination.

for all unsupervised methods. We include FISTA with the ground-truth dictionary as an upper bound: it shows what is achievable with a perfect dictionary and isolates dictionary learning as the variable under test. FISTA (oracle) remains near-perfect (MCC ≥ 0.95, OOD accuracy ≈ 0.97), confirming the problem is solvable at all scales. DL-FISTA degrades in MCC as d_z grows—dropping to ~0.3 by $d_z = 10$K—but maintains a small ID–OOD gap, indicating that its failure is dictionary quality, not compositional generalisation. When dictionary learning succeeds ($d_z \leq 500$), DL-FISTA dominates the linear probe on OOD accuracy (0.92 vs 0.83). SAE codes offer no consistent advantage over probing raw activations at any scale, consistent with Kantamneni et al. (2025). Varying sparsity k with d_z fixed yields the same pattern (appendix, Figure 16).

Takeaway. The oracle proves the problem is solvable at all scales—the OOD failure is not inherent to superposition. DL-FISTA beats linear probes when dictionary learning succeeds ($d_z \leq 500$) but falls behind at scale, confirming dictionary learning as the universal bottleneck. SAE codes offer no downstream advantage over raw activations, consistent with Kantamneni et al. (2025).

4.3 MORE DATA DOES NOT CLOSE THE AMORTISATION GAP

One issue with the evaluation setup could be that SAEs are simply data-limited: with enough training samples, the encoder should learn to approximate per-sample inference. If this were the case, the gap between SAEs and sparse coding

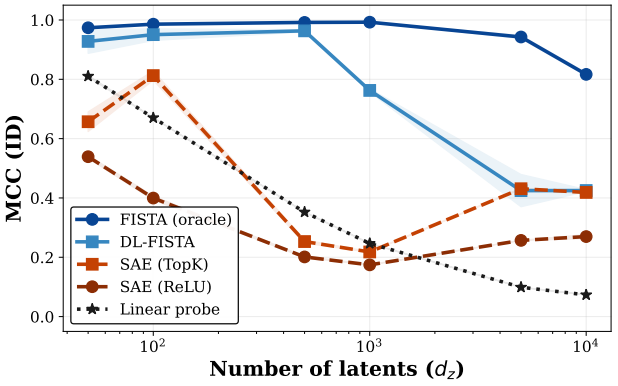

Figure 6: **Scaling latent dimension does not close the compositional gap.** DL-FISTA degrades due to dictionary learning difficulty but maintains a small ID–OOD gap. SAEs plateau at 0.2–0.5 MCC. The linear probe degrades sharply as superposition intensifies. $k = 10$, $p = 5000$, d_y follows the CS bound.

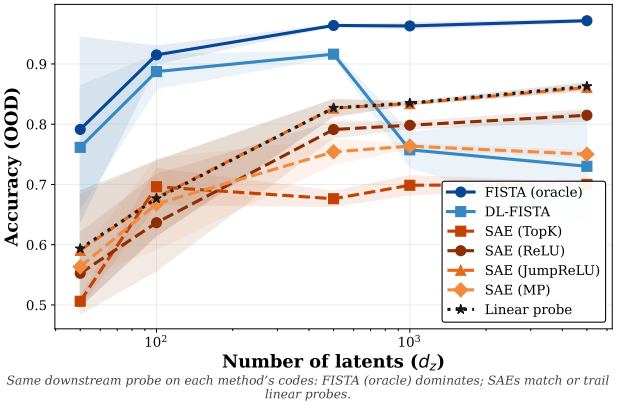

Figure 7: **DL-FISTA beats linear probes when dictionary learning succeeds but falls behind at scale.** OOD accuracy of a logistic probe trained on each method's codes. FISTA (oracle) dominates at all scales. SAE codes offer no advantage over probing raw activations. $k = 10$, $p = 5000$.

would shrink as the training set grows.

We vary the number of training samples $p \in \{10^2, \ldots, 10^5\}$ with all other parameters held fixed and report ID MCC (Figure 10) and OOD accuracy (Figure 11). FISTA (oracle) is constant by construction (it uses no training data beyond the oracle dictionary). DL-FISTA benefits substantially from more data—its MCC jumps from ~0.5 at $p = 10^2$ to ~0.98 by $p = 10^3$ and saturates—confirming that dictionary learning is genuinely sample-limited and that per-sample inference exploits a better dictionary immediately. SAE variants show a different pattern. SAE (TopK) and SAE (ReLU) improve only marginally, plateauing around 0.35–0.45 MCC. SAE (JumpReLU) *degrades* with more data, dropping from ~0.45 to below 0.1, suggesting that additional training leads to a poor local solution rather than correcting it. The gap between

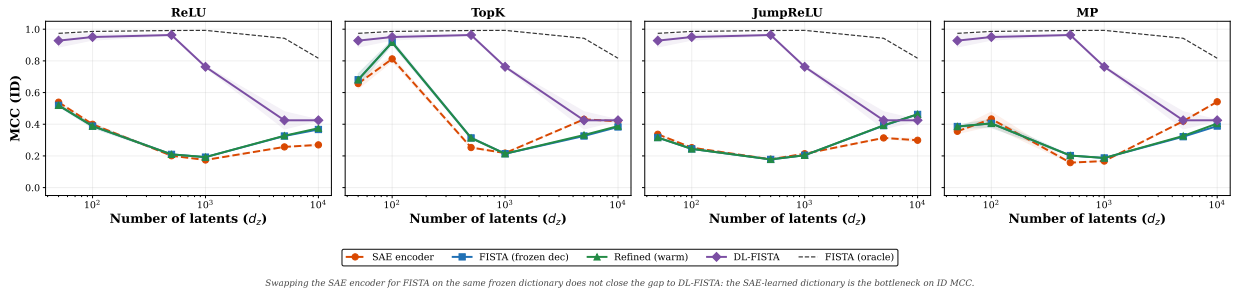

Figure 8: **The SAE-learned dictionary is the bottleneck, not the encoder.** Each panel shows one SAE type. Swapping the SAE encoder for FISTA on the same frozen dictionary (blue) or warm-starting from SAE codes (green) does not close the gap to DL-FISTA (purple), which learns its own dictionary. The oracle (gray dashed) confirms the problem is solvable. $k = 10$, $p = 5000$.

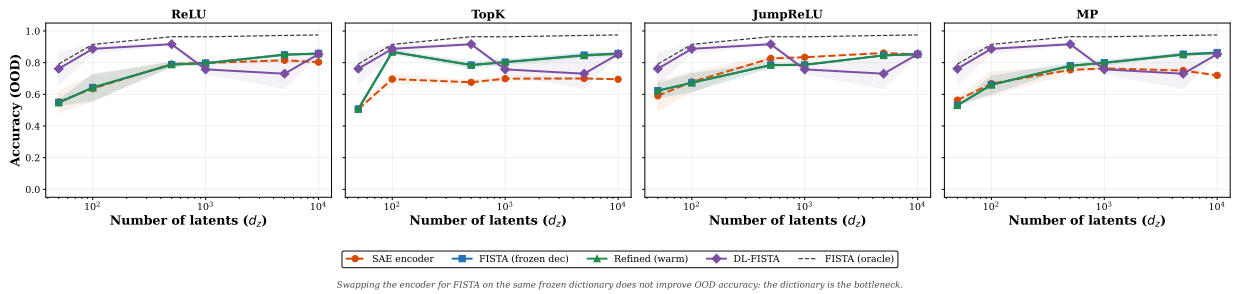

Figure 9: **Swapping the encoder does not improve OOD accuracy: the dictionary is the bottleneck.** Same layout as Figure 8. Replacing the SAE encoder with FISTA on the same frozen dictionary yields no consistent accuracy gain. The gap to DL-FISTA confirms that the dictionary directions, not the inference procedure, limit OOD performance. $k = 10$, $p = 5000$.

per-sample and amortised methods is stable or widening across two orders of magnitude of additional data. On the downstream prediction task (Figure 11), DL-FISTA clearly separates from the linear probe once $p \geq 10^3$ (0.88 vs 0.68 accuracy), while all SAE variants trail the linear probe—this is the regime ($d_z = 100$) where dictionary learning works, and it translates directly into better OOD generalisation. OOD AUC shows the same pattern (appendix, Figure 20).

Takeaway. Additional training data closes the dictionary learning gap (benefiting DL-FISTA) but does not close the amortisation gap. When dictionary learning succeeds, DL-FISTA dominates linear probes on the same downstream task, whereas SAE codes trail linear probes regardless of number of training samples.

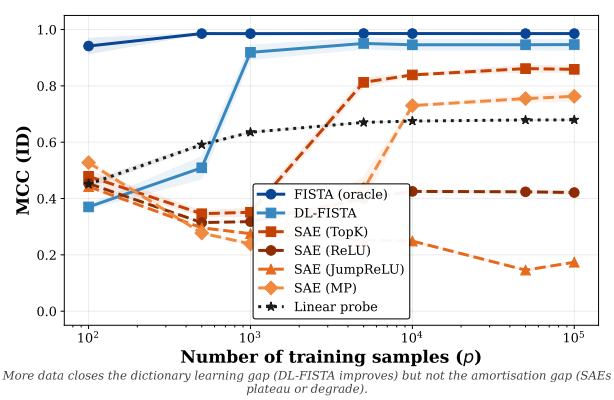

Figure 10: **More data closes the dictionary learning gap but not the amortisation gap.** DL-FISTA's MCC jumps from ~0.5 to ~0.98 by $p = 10^3$, confirming dictionary learning is sample-limited. SAEs plateau or degrade with more data, and the gap between per-sample and amortised methods is stable across two orders of magnitude. $d_z = 100$, $k = 10$, $d_y = 47$.

4.4 THE BOTTLENECK IS THE DICTIONARY, NOT THE ENCODER

The preceding experiments confound two potential sources of SAE failure: a poor dictionary and poor inference. We isolate these with controlled experiments that hold one component fixed while varying the other.

Swapping the encoder does not close the gap. Given a trained SAE with decoder $\hat{\mathbf{W}}$ and bias $\hat{\mathbf{b}}$, we compare three inference strategies on the same test input $\mathbf{y}$:

1. **SAE encoder** (baseline): a single forward pass, $\mathbf{h} = r(\mathbf{y})$.
2. **Frozen decoder**: FISTA on $\mathbf{y} - \hat{\mathbf{b}}$ using $\hat{\mathbf{W}}$ as a fixed